Dashboard Project Report

Submitted by:

Vikesh Gohil (IU2453000075)

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Acknowledgement

I would like to express my sincere gratitude to all those who have supported me throughout the development of this Dashboard. This project is the result of collective effort, guidance, and encouragement from many individuals.

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CERTIFICATE

This is to certify that the "**Crime Data Analysis Dashboard Web-application**" is the bonafide work carried out by **Vikesh Gohil (IU2453000075)**, a student of MCA SEM - III during the academic year 2025, in partial fulfillment of the requirements for the award of the degree of Master of Computer Application.

Internal Guide Signature :

Head of Dept. Signature :

Date :

Date :

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INTRODUCTION

The project titled "**Crime Data Analysis of Indian Cities**" is developed with the aim of understanding, analyzing, and predicting crime patterns across various cities in India. Crime data is a valuable resource for law enforcement agencies, policymakers, and the general public. However, in most cases, the data is large, complex, and difficult to interpret manually.

This project uses Data Analytics techniques to explore and visualize crime statistics in a meaningful way. It allows users to load crime data (from NCRB or other sources), apply filters based on year, state, city, and crime type, and view the data through interactive charts and graphs such as bar charts, line graphs, and heatmaps.

To enhance the analytical capability, the project also integrates a Machine Learning component using the Linear Regression algorithm. This model helps in predicting future crime trends based on historical data, providing insights like whether crime rates are increasing or decreasing over time.

The system is implemented using Python, with Streamlit as the front-end framework for building an interactive dashboard, and Pandas, Plotly, and Scikit-learn for data processing, visualization, and prediction. The project contributes to better decision-making and crime prevention by offering a data-driven view of public safety.

Existing System

i. Functional Description of the Existing System

- Currently, crime data in India is collected and maintained by various authorities such as the National Crime Records Bureau (NCRB) and local police departments.
- Most existing systems primarily store raw crime statistics in spreadsheets, PDFs, or basic databases.
- The data is often categorized by crime type, city, district, and year.
- Users can manually access reports to check crime trends but cannot perform interactive analysis or visualizations easily.
- Functionalities are limited to data retrieval and basic report generation; predictive analytics or trend identification is usually absent.

ii. Technical Architecture of the Existing System

- Data is usually stored in centralized databases maintained by government agencies.
- The system uses static web interfaces or offline reports for users to view the data.
- Analysis is mostly done using manual calculations in Excel or other spreadsheet tools.
- There is no real-time data integration, and updates occur periodically (annually or quarterly).
- Security is basic, with limited access control and no advanced data analytics infrastructure.

iii. Performance and Scalability

- Existing systems are not optimized for large-scale analysis or interactive querying.
- Performance is slow when handling large datasets spanning multiple years and cities.
- Scalability is limited due to manual processes and lack of automated data pipelines.
- Integration of new data sources (like social media crime reports, IoT devices, or citizen complaints) is difficult.

iv. Limitations of the Existing System

- Lack of data visualization and insights, making it hard for policymakers to identify patterns quickly.
- Manual processing leads to errors and delayed reporting.
- No support for predictive analysis or forecasting future crime trends.
- Difficult to compare crime across multiple cities or states effectively.
- Accessibility issues, as data is often in PDF or text formats that are not machine-readable.
- Limited interactivity—users cannot drill down into specific crime categories or perform customized queries.

Summary

The existing system focuses primarily on data storage and static reporting rather than analysis. It lacks interactive features, predictive analytics, and visualization capabilities. Performance, scalability, and user experience are limited due to manual processes and outdated architecture. These limitations highlight the need for an automated, interactive, and scalable crime data analysis system, which is the goal of our project.

Problem Areas of Existing System

1. Manual Data Processing:

Crime data is mostly handled manually using spreadsheets or text reports, which is time-consuming and error-prone.

2. Lack of Data Visualization:

Existing systems do not provide interactive charts, graphs, or dashboards, making it difficult to understand trends at a glance.

3. Limited Analytical Capabilities:

There is no predictive analysis or forecasting to identify future crime hotspots. Comparative analysis between cities or states is cumbersome.

4. Poor Performance with Large Data:

Handling multi-year crime data for multiple cities is slow and inefficient, affecting performance and user experience.

5. Scalability Issues:

The system cannot easily incorporate new data sources like online complaints, social media reports, or IoT-based alerts.

6. Accessibility Challenges:

Data is often in PDF or non-machine-readable formats, limiting easy access and automated processing.

7. Delayed Reporting:

Since data updates are periodic (annual/quarterly), there is no real-time or near real-time information, which affects timely decision-making.

8. Limited Security & Data Control:

Access control and data security are basic, increasing the risk of misuse or unauthorized access.

Need for the New System

1. Automated Data Processing:

To reduce manual work and errors by automatically collecting, storing, and organizing crime data from multiple sources.

2. Interactive Data Visualization:

To provide graphs, charts, and dashboards for better understanding of crime trends and patterns.

3. Predictive Analysis and Forecasting:

To enable identification of future crime hotspots and trends, helping authorities make proactive decisions.

4. Improved Performance and Scalability:

To handle large datasets spanning multiple years and cities efficiently, with faster data retrieval and processing.

5. Enhanced Accessibility:

To make crime data easily accessible online in machine-readable formats for researchers, policymakers, and citizens.

6. Real-Time or Near Real-Time Updates:

To allow timely analysis by incorporating latest crime reports and data automatically.

7. Advanced Security and Access Control:

To protect sensitive data with role-based access and secure data management.

8. Decision Support for Authorities:

To assist law enforcement and government agencies in strategic planning and crime prevention through actionable insights.

Proposed System

1. Overview:
- The proposed system is an automated crime data analysis platform designed to collect, process, and visualize crime data from Indian cities.
- It aims to provide interactive dashboards, trend analysis, and predictive insights to help authorities and researchers make informed decisions.

2. Functional Features:
- Data Collection: Automatically gathers crime data from sources like NCRB reports, local police databases, and online crime records.
- Data Storage: Stores data in a structured database (e.g., SQL/NoSQL) for efficient querying and analysis.
- Data Analysis: Supports statistical analysis, trend identification, and predictive modeling using tools like Python or R.
- Visualization: Interactive charts, graphs, heatmaps, and dashboards for easy understanding of crime patterns.
- User Roles: Different access levels for admin, law enforcement, and public users with secure authentication.
- Reporting: Generates custom reports and summaries based on selected cities, crime types, or years.

3. Technical Architecture:
- Front-end: Web interface using technologies like HTML, CSS, JavaScript, or React for interactive dashboards.
- Back-end: Server-side application (Python/Node.js) for processing data, handling requests, and managing databases.
- Database: Centralized storage using SQL/NoSQL databases for structured and scalable data management.
- Analytics Engine: Python/R modules for statistical analysis, trend forecasting, and predictive modeling.

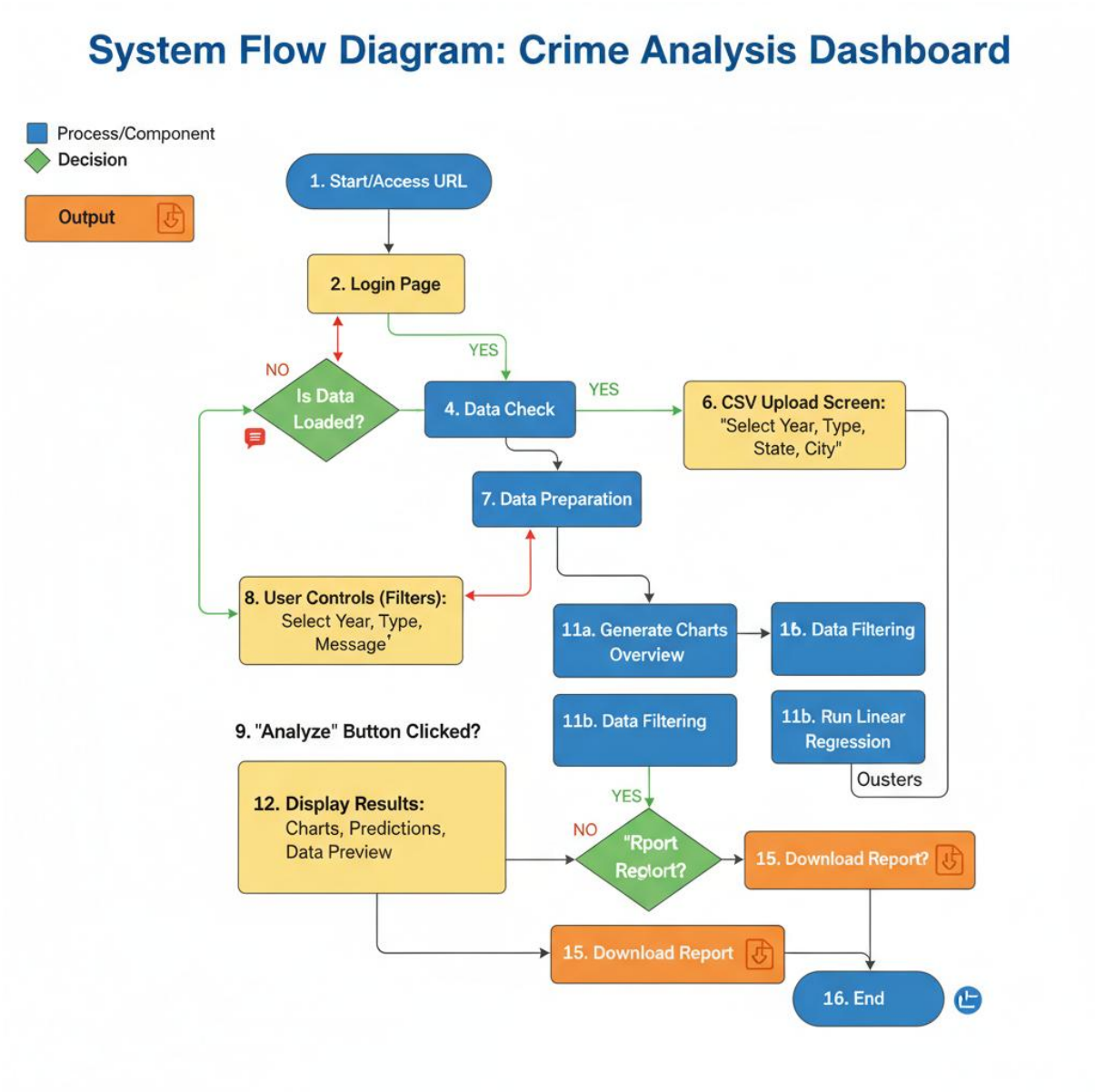
4. Advantages of the Proposed System:
- Automated and Efficient: Eliminates manual processing and reduces errors.
- Insightful Visualizations: Helps in quickly understanding crime trends and hotspots.
- Predictive Capabilities: Supports forecasting and proactive decision-making.

- Scalable and Extensible: Can handle large datasets and integrate new data sources in the future.
- Secure and Accessible: Role-based access ensures data security, and web-based access enables easy usability.

5. Expected Outcome:

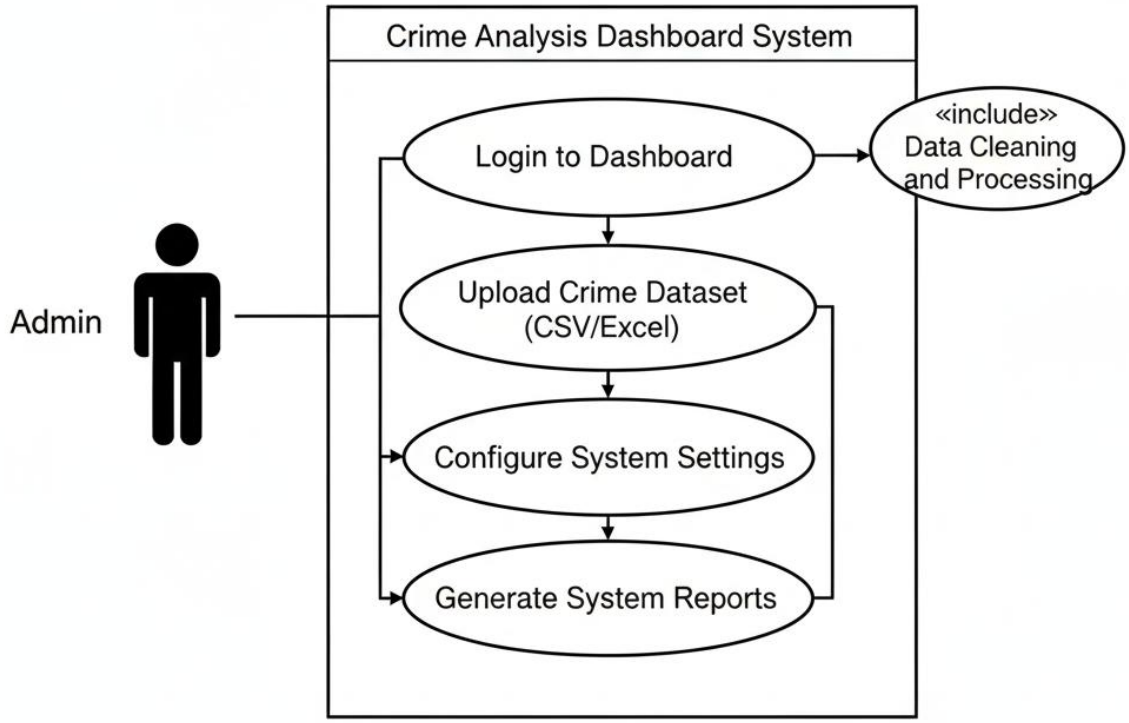
- A user-friendly, interactive platform for crime data analysis.
- Faster insights for policymakers and law enforcement agencies.
- Improved public awareness through accessible crime statistics and trends.

System Flow Diagram

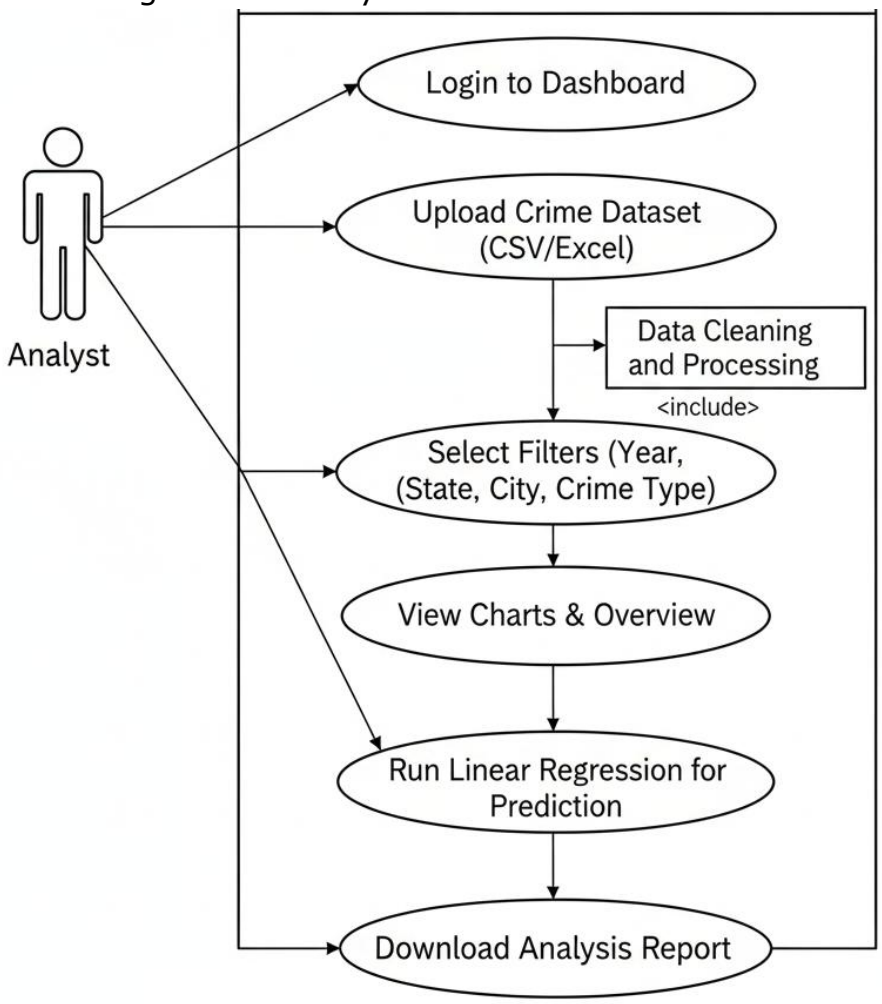


UML

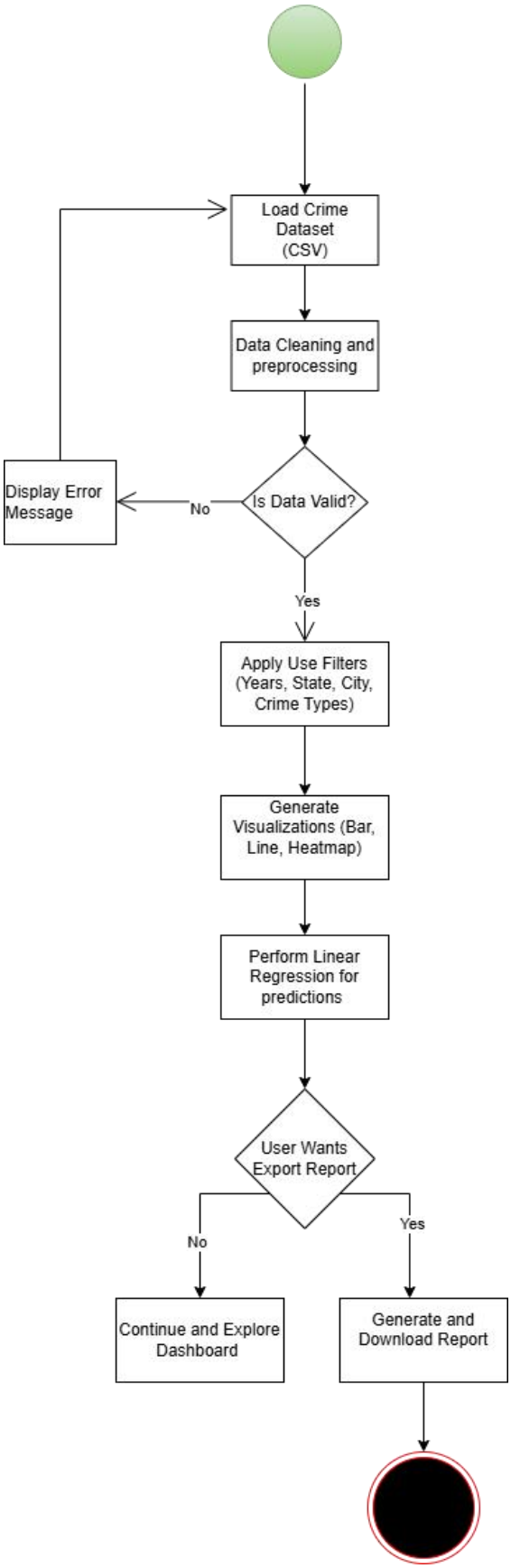
Use case diagram for Admin role



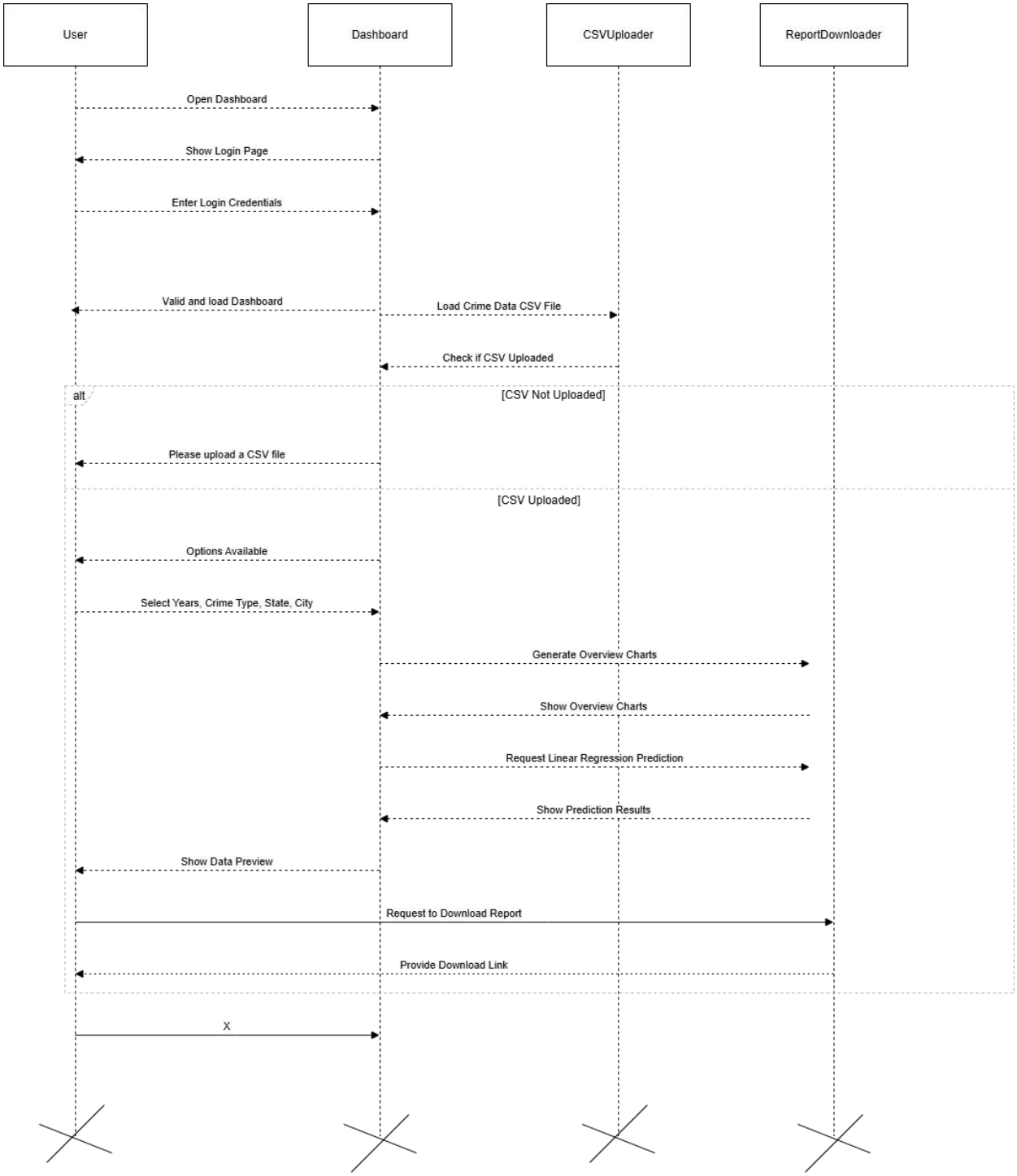
Use case diagram for Analyst role



Activity-Diagram

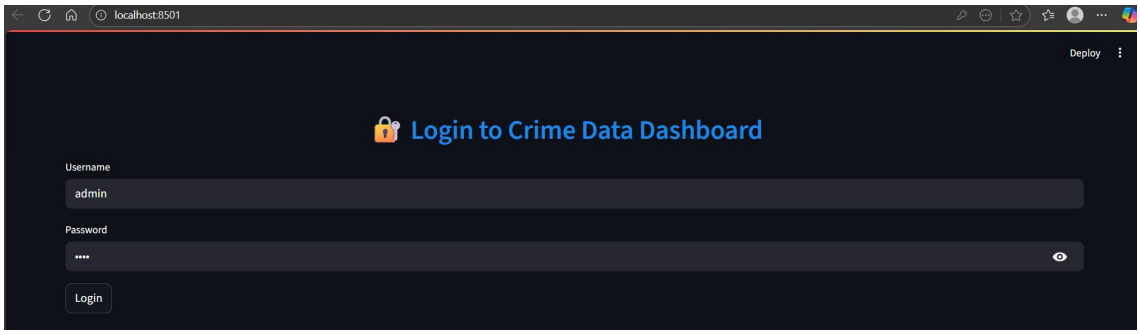


Sequence Diagram

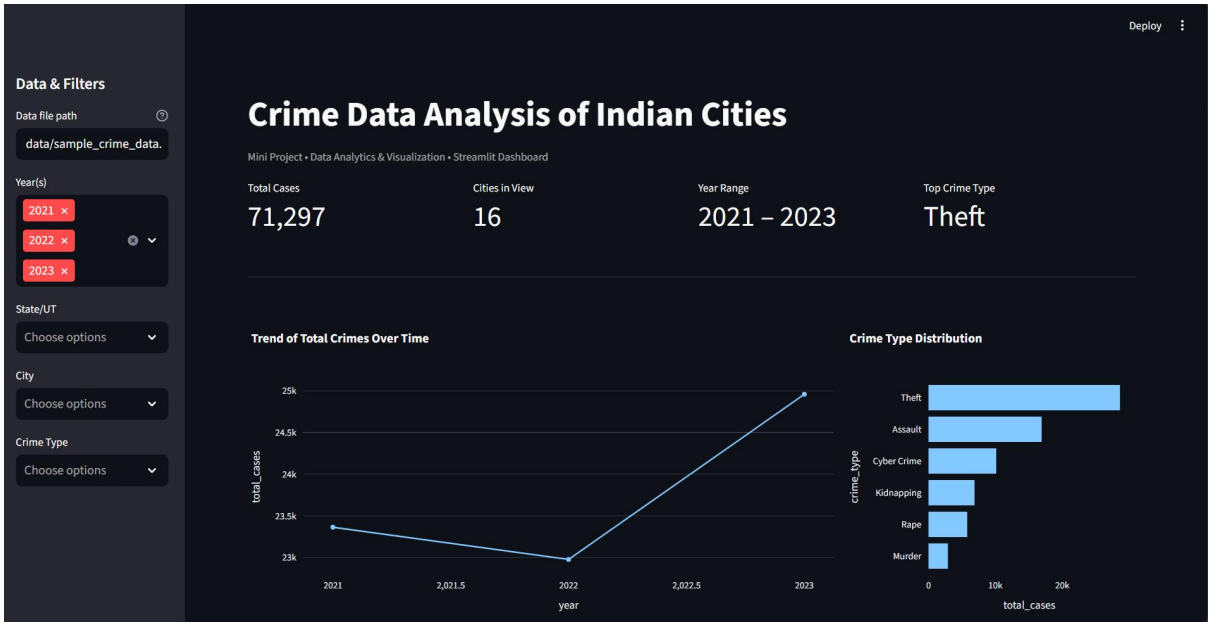


User Interface

1. Login Page



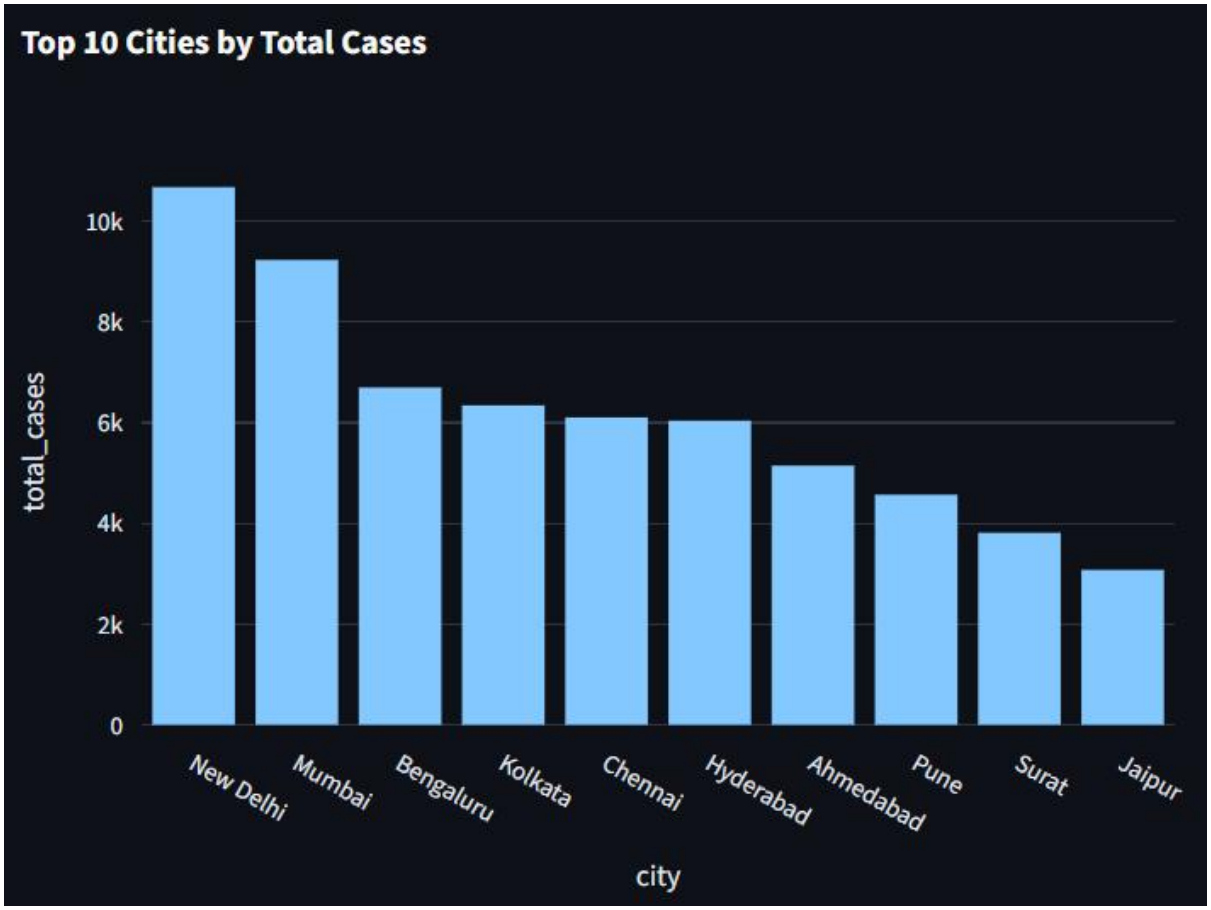
2. Dashboard Overview



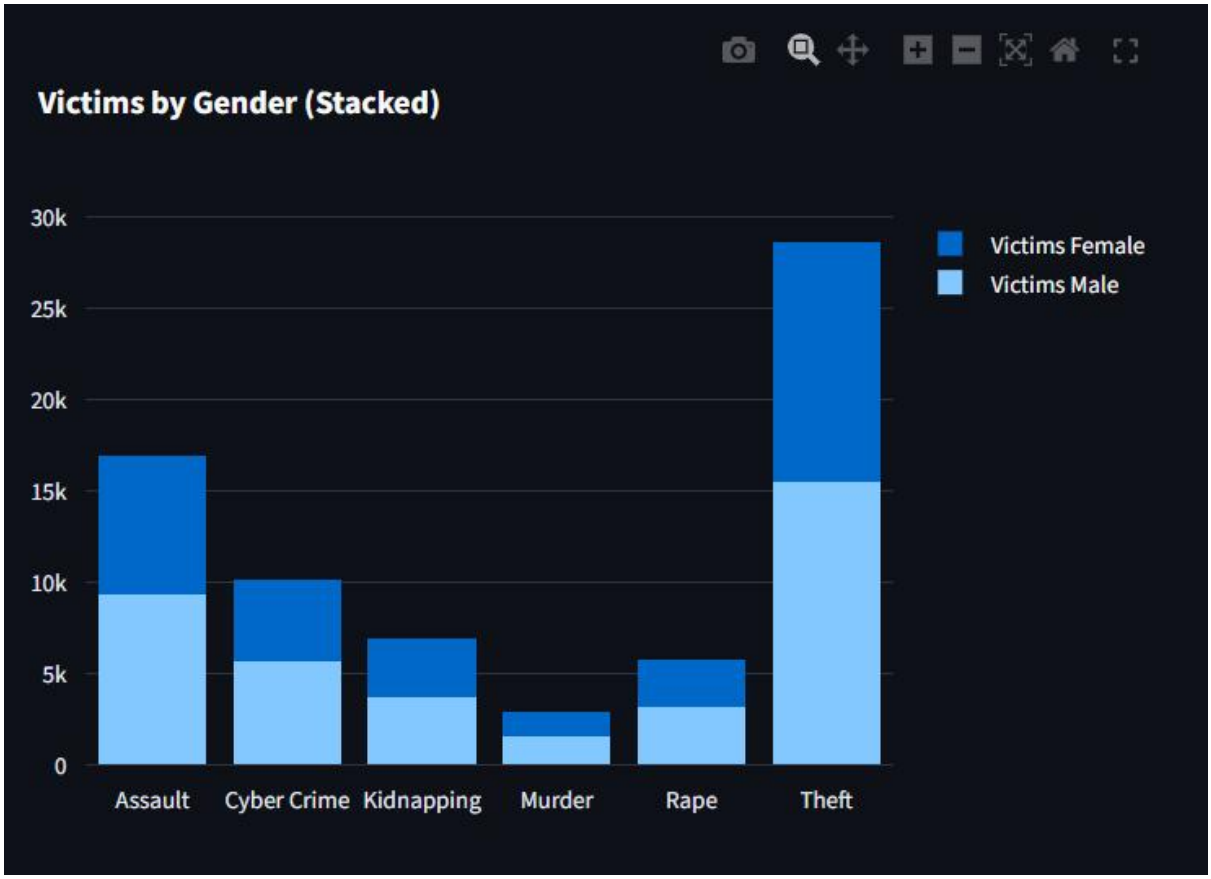
3. Different Charts

4. Feature Prediction

5. Bar Chart for different City



6. Stacked Chart



7. Data Preview as per Filter

	state_ut	city	year	crime_type	victims_male	victims_female	total_cases
108	Gujarat	Ahmedabad	2021	Theft	367	360	727
109	Gujarat	Ahmedabad	2021	Assault	157	190	347
110	Gujarat	Ahmedabad	2021	Kidnapping	88	51	139
111	Gujarat	Ahmedabad	2021	Rape	90	52	142
112	Gujarat	Ahmedabad	2021	Murder	28	36	64
113	Gujarat	Ahmedabad	2021	Cyber Crime	163	94	257
114	Gujarat	Ahmedabad	2022	Theft	481	268	749
115	Gujarat	Ahmedabad	2022	Assault	257	158	415
116	Gujarat	Ahmedabad	2022	Kidnapping	94	58	152
117	Gujarat	Ahmedabad	2022	Rape	55	53	108

Test Plan and Test Cases

1. Test Plan

1.1 Objective

- The objective of this test plan is to verify that the Crime Data Analysis system works as expected by:
- Correctly processing and analyzing crime data from multiple cities.
- Generating accurate visual reports and charts.
- Allowing users to download reports.
- Ensuring usability and system reliability.

1.2 Scope

- In Scope:
  - Data import and validation.
  - Data analysis (city-wise, crime-type-wise, year-wise).
  - Visualization (charts, graphs).
  - Report generation and download functionality.

- Out of Scope:
  - Real-time crime reporting.
  - User authentication (if not included in your system).

1.3 Test Strategy

- Functional Testing: Check all features work as intended.
- Usability Testing: Ensure the system is user-friendly.
- Data Accuracy Testing: Validate that analytics and reports are correct.
- Performance Testing: Ensure the system handles large datasets efficiently.

1.4 Test Environment

- Front-end: HTML, CSS, JavaScript
- Back-end: Python (Pandas, Matplotlib, or Django/Flask if web-based)
- Database: CSV / SQL / Excel data files
- Tools: Jupyter Notebook, Browser (Chrome/Edge)

1.5 Deliverables

- Test Plan Document
- Test Cases Document
- Test Summary Report

2. Test Cases

Test Case ID	Test Scenario	Test Steps	Expected Result	Status (Pass/Fail)
TC_01	Import crime data	1. Upload CSV/Excel file with crime data	Data should be imported without errors	Pass
TC_02	Validate data format	1. Check if all required columns exist (City, Crime Type, Year, Cases)	Missing columns should show an error message	Pass
TC_03	Analyze city-wise crime data	1. Select a city from the dropdown 2. Generate analysis	Display correct crime statistics for the selected city	Pass
TC_04	Analyze crime-type-wise data	1. Select a crime type 2. Generate report	Report shows correct number of cases per year/city for that crime type	Pass
TC_05	Generate visual chart	1. Click on "Generate Chart"	Correct chart is displayed (bar, line, or pie chart)	Pass

Test Case ID	Test Scenario	Test Steps	Expected Result	Status (Pass/Fail)
TC_06	Download report	1. Click on "Download Report"	Report is downloaded in PDF/Excel format	Pass
TC_07	Handle empty data file	1. Upload empty CSV/Excel	System shows error message "No data found"	Pass
TC_08	Performance with large data	1. Upload dataset with 10000+ records	System should process data within reasonable time (e.g., <5s)	Fail
TC_09	Invalid file format	1. Upload a non-CSV/Excel file	System shows error message "Invalid file format"	Fail
TC_10	Check UI responsiveness	1. Open system in different screen sizes	Layout should adjust and display correctly	Pass

Future Enhancements

These are potential improvements and advanced features that could be added to your project in the future to make it more powerful, interactive, and insightful:

1. Advanced Analytics

- Implement predictive analytics using machine learning to forecast crime trends in different cities.
- Detect crime hotspots using geospatial analysis and heatmaps.
- Identify patterns like crime type frequency over time or correlation between crime and other socio-economic factors.
- Use interactive tools like Power BI, Tableau, or Dash to create dashboards.

2. Real-Time Data Integration

- Connect the system with police databases or public crime data APIs for real-time updates.
- Enable automatic data updates without manual uploads.

3. User Management and Role-Based Access

- Implement login-based user roles: Admin, Analyst, Viewer.
- Restrict access to sensitive data for specific users.

4. Report Customization

- Allow users to select multiple cities and crime types for combined reports.
- Provide multiple file format options for downloading reports: PDF, Excel, CSV.
- Include visual report customization: colors, chart types, and layouts.

6. Alerts and Notifications

- Send email alerts or dashboard notifications for spikes in crime rates.
- Enable users to subscribe to city-specific crime alerts.

7. Data Quality and Cleaning Automation

- Implement automatic detection of missing or inconsistent data.
- Include data normalization for consistent reporting across cities.

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